PES University - Dept. of CSE
Lab 1: Requirements Engineering & UML Use-Case Modelling
Problem Statement #43 - Feature Flag & Dynamic Config Manager

Complete Requirements Table

Target Stakeholders / Actors: Software Engineer, Release Manager (primary); Client SDK (secondary / system actor).

Functional Requirements (FR-001 to FR-005)

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall evaluate user targeting rules (e.g., User ID hash, Beta cohort) and return accurate boolean feature flag states.	High	Pass: Target 10% rollout cohort receives enabled flag; Fail: User outside cohort receives enabled flag.	Core evaluation logic; every other feature (rollout, override, sync) depends on this returning a correct result.
FR-002	Functional	The system shall support creation of feature flags with a unique key, description, and default state.	High	Pass: Duplicate key creation is rejected; new flag defaults to off.	A unique key and safe default prevent naming collisions and accidental exposure of unfinished features.
FR-003	Functional	The system shall support percentage-based rollout configuration per user cohort.	High	Pass: Changing rollout percentage takes effect on the next evaluation cycle.	Percentage rollout is the primary mechanism for gradual, risk-controlled feature releases.
FR-004	Functional	The system shall support environment-specific configuration overrides (dev/staging/production).	Medium	Pass: A production override does not affect staging evaluation of the same flag.	Environments must be isolated so a change tested in staging cannot unintentionally reach production.
FR-005	Functional	The system shall synchronize flag state changes to connected client SDKs in real time.	High	Pass: SDK reflects the updated state within 2 seconds without a client restart.	Keeps distributed clients consistent and avoids stale flag states causing inconsistent user experience.

Non-Functional Requirements (NFR-001 & NFR-002)

ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Performance & Security	The feature flag evaluation API must execute in under 15 ms to minimize overhead when integrated into client critical paths.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.	Flag evaluation sits on the hot path of client requests, so added latency directly affects

					application performance.
NFR-002	Availability	The flag evaluation service must degrade gracefully by serving the last-known-good configuration if the backend is unreachable.	High	Pass: SDK continues serving cached flags during a simulated backend outage.	Feature flags often gate critical functionality; the evaluation service failing must not take down dependent applications.